

INTERNSHIP PROJECT REPORT

EMPLOYEE ATTRITION PREDICTION DASHBOARD

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1. Introduction

Employee attrition refers to employees leaving an organization through resignation, retirement, or other forms of separation. Understanding patterns associated with attrition can help HR teams identify areas that may require attention and support data-informed workforce planning.

This project analyzes the IBM HR Analytics Employee Attrition & Performance dataset and develops an employee attrition analysis workflow using Python, machine learning, and Power BI. The project combines exploratory data analysis with a Logistic Regression model and an interactive dashboard.

2. Project Objectives

- Clean and prepare the employee dataset for analysis.
- Explore employee characteristics and their association with attrition.
- Build a machine learning model to predict employee attrition.
- Evaluate the model using accuracy, precision, recall, F1-score, and a confusion matrix.
- Create a Power BI dashboard that communicates major attrition patterns and key metrics.

3. Dataset Description

The project uses the IBM HR Analytics Employee Attrition & Performance dataset. The original dataset contains 1,470 employee records and 35 columns. Four administrative or non-informative columns were removed during cleaning: EmployeeCount, EmployeeNumber, Over18, and StandardHours. The resulting working dataset contains 1,470 rows and 31 columns.

Item	Value
Records	1,470
Original columns	35
Working columns	31
Missing values	0
Duplicate rows	0
Employees who left	237
Employees who stayed	1,233
Overall attrition rate	16.12%

4. Data Cleaning and Preparation

- Loaded the Excel dataset into Python using pandas.
- Removed four administrative columns that were not useful for modeling.
- Checked for missing values; none were found.
- Checked for duplicate rows; none were found.
- Converted the Attrition target from Yes/No to 1/0 for machine learning.
- Separated the target variable (Attrition) from the predictor variables.
- Used a stratified 80/20 train-test split to preserve the class distribution.
- Applied one-hot encoding to categorical variables using a preprocessing pipeline.

5. Exploratory Data Analysis

The exploratory analysis focused on identifying patterns associated with employee attrition. These results describe associations in the dataset and should not be interpreted as proof that a particular factor causes employees to leave.

Factor	Observed attrition pattern
Department	Sales: 20.63%; HR: 19.05%; R&D: 13.84%
Overtime	OverTime = Yes: 30.53%; OverTime = No: 10.44%
Job Satisfaction	Level 1: 22.84%; Level 4: 11.33%
Monthly Income	₹1,009–₹5,000: 21.76%; ₹15,001–₹19,999: 3.76%
Age	18–25: 35.77%; 26–35: 19.14%; 36–45: 9.19%
Business Travel	Travel_Frequently: about 24.9%; Non-Travel: about 8%
Marital Status	Single: about 25.5%; Married: about 12%; Divorced: about 10%
Job Role	Sales Representative: about 39.7%; Research Director: about 2.5%

6. Machine Learning Model

A Logistic Regression classifier was selected as a baseline model because it is relatively simple, interpretable, and suitable for binary classification. A preprocessing pipeline was used to encode categorical variables while passing numerical variables through the model.

Train-test split:

- Training set: 1,176 records
- Testing set: 294 records
- Test size: 20%
- Random state: 42
- Stratified split: Yes

6.1 Baseline Model Results

The initial Logistic Regression model achieved 87.75% accuracy. However, because attrition is the minority class, accuracy alone did not provide a complete picture of performance.

Class	Precision	Recall / F1
Stayed (0)	0.99	Recall 1.00; F1 0.93
Left (1)	0.92	Recall 0.26; F1 0.40

Confusion matrix: $\begin{bmatrix} 246 & 1 \\ 35 & 12 \end{bmatrix}$. The model identified only 12 of the 47 employees who left in the test set, illustrating the effect of class imbalance.

6.2 Balanced Model Results

To give the minority attrition class greater importance, `class_weight='balanced'` was applied to Logistic Regression. This reduced overall accuracy but improved recall for the attrition class.

Metric	Stayed (0)	Left (1)	Overall
Precision	0.92	0.35	Macro 0.63
Recall	0.77	0.64	Macro 0.71
F1-score	0.84	0.45	Macro 0.65
Accuracy	-	-	0.75

Confusion matrix: $\begin{bmatrix} 191, & 56 \\ 17, & 30 \end{bmatrix}$. The balanced model detected 30 of the 47 attrition cases in the test set, giving an attrition recall of 64%. The trade-off is lower overall accuracy compared with the unbalanced baseline.

7. Power BI Dashboard

An interactive Power BI dashboard was created to communicate the project's main findings. The dashboard contains three KPI cards and six analytical charts.

- Total Employees: 1,470
- Employees Left: 237
- Attrition Rate: 16.12%
- Attrition by Department
- Attrition by Job Role
- Attrition by Overtime
- Attrition by Business Travel
- Attrition by Marital Status
- Attrition by Education Field



Figure 1. Employee Attrition Prediction Dashboard

8. Tools and Technologies Used

Tool / Technology	Purpose
Python	Data analysis and machine learning
Pandas	Data loading and manipulation
NumPy	Numerical operations
Scikit-learn	Preprocessing, train-test split, and Logistic Regression
Joblib	Saving the trained model
Matplotlib / Seaborn	Exploratory data visualization
Power BI	Interactive dashboard development
Excel	Dataset storage and preparation

9. Key Findings

- The overall attrition rate in the dataset is 16.12%.
- Employees working overtime had a higher observed attrition rate than employees who did not work overtime.
- The youngest age groups showed higher observed attrition rates than most older age groups in this dataset.
- Sales and HR showed higher observed department-level attrition rates than R&D.
- Lower monthly-income groups showed higher observed attrition rates than the highest income group in this dataset.
- The balanced Logistic Regression model improved detection of employees who left, with 64% recall for the attrition class on the test set.

10. Limitations

- The dataset is a fixed historical dataset and may not represent every organization or current workforce.
- Observed associations do not establish causation.
- The Logistic Regression model is a baseline model and could be compared with additional algorithms in future work.
- The attrition class is imbalanced, so accuracy should be interpreted together with recall, precision, and F1-score.
- The Power BI dashboard currently presents analytical results; the saved machine-learning model is maintained separately in Python.

11. Conclusion

This project developed an end-to-end employee attrition analysis workflow using data cleaning, exploratory analysis, machine learning, and business intelligence visualization. The analysis identified several employee characteristics associated with different observed attrition levels, while the Logistic Regression model demonstrated the importance of evaluating minority-class recall rather than relying only on overall accuracy. The Power BI dashboard provides a concise visual summary of the dataset and can support further HR analysis and discussion.

12. Future Scope

- Compare Logistic Regression with tree-based models such as Random Forest or Gradient Boosting.
- Tune classification thresholds to balance false positives and false negatives.
- Add model probability outputs to an operational dashboard.
- Use newer organizational data to monitor whether attrition patterns change over time.
- Add interactive slicers for department, job role, age group, and other employee attributes.

End of Report